

Chapter 4

Artifact: approximation under (A1) and (A6)

Neural network values converge to π

The finite-space result of Bühler et al. fails on a general probability space at (F1) and (F3). The tools of this section repair both under the Chapter 3 hypotheses (A1) ($Z \in L^\infty$) and (A6) ($W(\delta) \geq 0$ a.s. for every $\delta \in \mathcal{H}$): Lemma 4.6 restores a bounded near-optimal target, and Lemma 4.4 restores the limit passage inside the OCE. The following theorem combines them.

Theorem 4.1. *Assume the standing market and network setting of Bühler et al., let $\rho = \rho_\ell$ be an OCE risk measure, and assume (A1) and (A6). Then*

$$\lim_{M \rightarrow \infty} \pi_M(-Z) = \pi(-Z)$$

in $[-\infty, \infty]$, the convergence being monotone from above.

Proof. Write $X := -Z$. We distinguish the three cases $\pi(X) = +\infty$, $\pi(X) \in \mathbb{R}$ and $\pi(X) = -\infty$.

Case $\pi(X) = +\infty$. By Lemma 4.1, $\pi_M(X) \geq \pi(X) = +\infty$ for every M , so $\pi_M(X) = +\infty$ for all M and the claim is trivial.

Case $\pi(X) \in \mathbb{R}$. By Lemma 4.1 it is enough to show that for every $\varepsilon > 0$ there exists M with $\pi_M(X) \leq \pi(X) + \varepsilon$.

Step 1: a bounded near-optimal target. By Lemma 4.6 there exists a bounded strategy $\delta \in \mathcal{H}^u$, say $|\delta_k| \leq j$ a.s. for all k , such that

$$\rho_\ell(X + W(H \circ \delta)) \leq \pi(X) + \frac{\varepsilon}{2}.$$

Write $\eta := H \circ \delta$ and $Y := X + W(\eta)$.

Step 2: representing maps and integrability. Each δ_k is $\mathcal{F}_k$ -measurable with $\mathcal{F}_k = \sigma(I_0, \dots, I_k)$, so there exist Borel maps $f_k : \mathbb{R}^{r(k+1)} \rightarrow \mathbb{R}^d$ with $\delta_k = f_k(I_0, \dots, I_k)$. Truncating each component of f_k to the interval $[-j, j]$

changes nothing $\mathbb{P}$ -a.s. and makes f_k bounded everywhere. In particular every component of f_k lies in $L^1(\mu_k)$, where μ_k is the law of $(I_0, \dots, I_k)$. This settles (F1).

Step 3: network approximation. By Hornik's theorem applied componentwise, there exist networks $F_{k,m} \in \mathcal{NN}_{\infty, r(k+1), d}$ with $F_{k,m} \rightarrow f_k$ in $L^1(\mu_k)$ as $m \rightarrow \infty$. Set

$$\delta^m := (F_{k,m}(I_0, \dots, I_k))_{k=0, \dots, n-1}.$$

Then $\delta_k^m \rightarrow \delta_k$ in $L^1(\mathbb{P})$ for every k . Passing to a subsequence (not relabelled) we may assume

$$\delta_k^m \rightarrow \delta_k \text{ } \mathbb{P}\text{-a.s.} \quad \text{for all } k = 0, \dots, n-1$$

simultaneously. Since each $F_{k,m}$ belongs to $\mathcal{NN}_{\infty, r(k+1), d}$ and these families are nested with $\bigcup_M \mathcal{NN}_{M, r(k+1), d} = \mathcal{NN}_{\infty, r(k+1), d}$, there exist $M_m \uparrow \infty$ such that $\delta^m \in \mathcal{H}_{M_m}$.

Step 4: constrained strategies and objectives. Set $\eta^m := H \circ \delta^m$. Continuity of each $H_k(\omega, \cdot)$ and induction over k give $\eta_k^m \rightarrow \eta_k$ $\mathbb{P}$ -a.s. for every k . Gains converge: $(\eta^m \cdot S)_T \rightarrow (\eta \cdot S)_T$ a.s. Upper semicontinuity of the $c_k(\omega, \cdot)$ yields

$$\limsup_{m \rightarrow \infty} C_T(\eta^m) \leq C_T(\eta) \text{ } \mathbb{P}\text{-a.s.},$$

and therefore, writing $Y_m := X + W(\eta^m)$,

$$\liminf_{m \rightarrow \infty} Y_m \geq Y \text{ } \mathbb{P}\text{-a.s.}.$$

Furthermore, by (A6) and (A1), $Y_m \geq -\|Z\|_\infty$ a.s. for every m .

Step 5: conclusion. Now we fulfill all the conditions to use Lemma 4.4, which gives $\limsup_m \rho_\ell(Y_m) \leq \rho_\ell(Y) \leq \pi(X) + \varepsilon/2$. Finally, choose m_0 with $\rho_\ell(Y_{m_0}) \leq \pi(X) + \varepsilon$. Since $\delta^{m_0} \in \mathcal{H}_{M_{m_0}}$, one has $\pi_M(X) \leq \pi(X) + \varepsilon$ for all $M \geq M_{m_0}$.

Case $\pi(X) = -\infty$. Let $L \in \mathbb{R}$ be arbitrary. Choose $\delta \in \mathcal{H}^u$ with $\rho_\ell(X + W(H \circ \delta)) \leq L - 1$. Then, steps 1 to 5 above go through with the threshold L in place of $\pi(X) + \varepsilon$, yielding M with $\pi_M(X) \leq L$. By choosing L lower and lower, and being $(\pi_M(X))_{M \in \mathbb{N}}$ non-increasing, $\pi_M(X) \rightarrow -\infty = \pi(X)$. $\square$